Chapter 10. Structure Learning: PC, GES and MMHC

Graphical models, Lecture 3 (constraint-based and score-based discovery of the Markov equivalence class)

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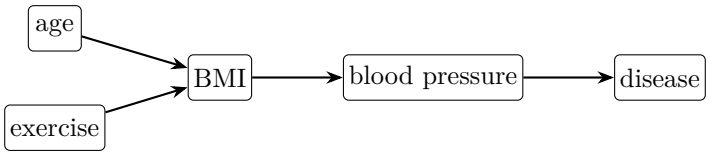
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1 What Observational Data Can Reveal

Which relations remain given the others, which edges do CIs remove, and how far can directions be fixed?



Structure learning returns the relations the assumptions and data allow; not every arrow is fixed.

2 Assumptions Matter as Much as Data

Assumption	Practical meaning
acyclicity	relations at one time point are represented by a DAG without cycles
causal Markov condition	paths blocked in the graph appear as conditional independences in the data
faithfulness	no accidental cancellation of coefficients makes relations look independent
causal sufficiency	no unobserved common cause of two analyzed variables
same generating process	all samples come from the same system, without selection bias

PC and GES need causal sufficiency; with hidden common causes use FCI (Ch. 11).

3 The PC Algorithm

Motivation. PC removes edges by CI tests with growing conditioning sets drawn from current neighbours, then orients from colliders.

3.1 Skeleton phase

Algorithm: PC, skeleton

1. Connect all pairs of variables by undirected edges.
2. For $\ell = 0, 1, 2, \ldots$: for each adjacent pair (X_j, X_k) and each conditioning set $S \subseteq \text{adj}(X_j) \setminus \{X_k\}$ of size ℓ, test
$$H_0 : X_j \perp\!\!\!\perp X_k \mid X_S.$$
3. If H_0 is not rejected for some S, remove the edge $X_j - X_k$,
4. and store that S as the separating set $\text{Sep}(X_j, X_k)$.

Stop when no node has more than ℓ remaining neighbors.

Tests: Fisher-z partial correlation (Gaussian, Ch. 8), χ^2/G^2 (categorical), nonlinear CI tests (Ch. 11). The output is an undirected **skeleton**.

Example: skeleton of the toy example

Pair	Conditioning set	Decision	Action
age, exercise	$\varnothing$	independent	remove edge
age, blood pressure	BMI	conditionally independent	remove edge
exercise, blood pressure	BMI	conditionally independent	remove edge
BMI, disease	blood pressure	conditionally independent	remove edge

(Age–disease, exercise–disease are removed similarly.) Remaining skeleton:

$$\mathcal{E}_{\text{skel}} = \{\text{age–BMI},\ \text{exercise–BMI},\ \text{BMI–blood pressure},\ \text{blood pressure–disease}\}.$$

3.2 Orientation phase

$\text{Sep}(\text{age}, \text{exercise}) = \varnothing$, both are adjacent to BMI and not to each other, and $\text{BMI} \notin \text{Sep}$, so

$$\text{age} \to \text{BMI} \leftarrow \text{exercise}.$$

Algorithm: PC, orientation

1. **v-structures**: for every unshielded triple $X - Z - Y$ (X, Y non-adjacent) with $Z \notin \text{Sep}(X, Y)$, orient $X \to Z \leftarrow Y$.
2. **Meek rules**, applied until nothing changes:

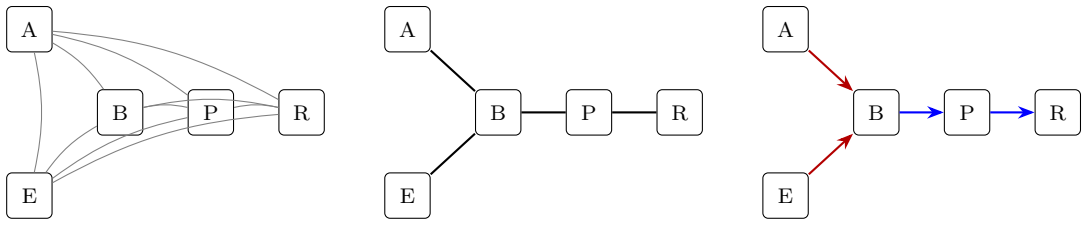
   R1 (no new collider) $X \to Z - Y$, X, Y non-adjacent $\Rightarrow Z \to Y$;

   R2 (no cycle) $X \to Z \to Y$ and $X - Y \Rightarrow X \to Y$;

   R3 $X - Z_1 \to Y$, $X - Z_2 \to Y$, $X - Y$, Z_1, Z_2 non-adjacent $\Rightarrow X \to Y$.

R1: $A \to B - C$, $A \not\sim C$; $C \to B$ would add an unsupported collider, so $B \to C$. R2: $A \to B \to C$, $A - C$; $C \to A$ would make a cycle, so $A \to C$. Example:

$$\text{age} \to \text{BMI} \leftarrow \text{exercise}, \qquad \text{BMI} \to \text{blood pressure} \to \text{disease}.$$



(1) complete graph (2) skeleton after CI tests (3) v-structure + Meek R1

Figure 1: PC on the toy example: CI tests give the skeleton; the v-structure $A \to B \leftarrow E$ (red) is oriented and R1 propagates $B \to P \to R$ (blue).

3.3 Why PC is correct

Theorem: correctness of PC with a CI oracle

Assume the Markov condition, faithfulness and causal sufficiency, and that every CI query is answered correctly. Then PC returns the CPDAG of the true DAG.

Proof [Supplement]. *Adjacencies.* By faithfulness adjacent variables are dependent given every set, so true edges stay. If X, Y are non-adjacent with Y not a descendant of X, $X \perp\!\!\!\perp Y \mid \mathrm{pa}(X)$ (local Markov); as $\mathrm{pa}(X) \subseteq \mathrm{adj}(X)$ throughout, a separating set is found by level $\ell = |\mathrm{pa}(X)|$. *v-structures.* For unshielded $X - Z - Y$: if Z is a non-collider, every separating set contains Z; if a collider, none does. So $\mathrm{Sep}(X, Y)$ decides correctly. *Remaining edges.* The Meek rules are sound (alternatives create a v-structure or cycle) and complete (Meek, 1995). ■

Remark: finite-sample issues

Early type-II errors remove edges irrecoverably and corrupt separating sets. *PC-stable* removes order dependence by updating adjacencies after each level. Conflicting v-structures signal test errors or violated assumptions.

4 Score-Based Search

4.1 BIC for Gaussian linear SEMs

$$S_{\mathrm{BIC}}(G) = \ell(\hat{\theta}_G) - \frac{d_G}{2} \log n,$$

with d_G the number of parameters.

Theorem: BIC decomposes into local scores

For a Gaussian linear SEM,

$$S_{\mathrm{BIC}}(G) = \sum_{j=1}^{p} S_j\{X_j, \mathrm{pa}_G(j)\}, \qquad S_j = -\frac{n}{2} \log \hat{\sigma}^2_{j|\mathrm{pa}(j)} - \frac{|\mathrm{pa}(j)| + 1}{2} \log n + c,$$

where $\hat{\sigma}^2_{j|\mathrm{pa}(j)}$ is the residual variance of the least-squares regression of X_j on its parents.

Proof [Supplement]. By the Markov factorization the log-likelihood is a sum over local regressions with separate parameters; each has maximum $-\frac{n}{2}\log(2\pi\hat{\sigma}^2) - \frac{n}{2}$ and $|\mathrm{pa}(j)|+1$ parameters (centered data). ■

A local move changes one or two local scores. Equivalent DAGs share maximum likelihood and d_G, so BIC is **score equivalent**: scores cannot split an equivalence class either.

4.2 GES and hill climbing

	GES	Hill climbing
search space	CPDAGs representing Markov equivalence classes	individual DAGs
moves	forward phase adds relations, backward phase removes them	add, delete or reverse an edge
typical output	a high-scoring CPDAG	one high-scoring DAG
caution	remaining undirected edges are undetermined	not every output direction is identified

Algorithm: Greedy Equivalence Search

Forward: from the empty graph, apply the best single edge insertion (in equivalence-class space) while the score increases. **Backward**: apply the best deletion while it helps. Output the CPDAG.

Theorem: consistency of GES (Chickering, 2002)

If the distribution is faithful to a DAG and the score is consistent and decomposable (e.g. BIC), then GES returns the true CPDAG with probability tending to 1 as $n \to \infty$, despite being greedy.

Hill climbing can stick in local optima (tabu lists, restarts help).

5 Computational Burden and Hybrid Methods

	Why the cost grows	Remaining problem
PC	repeats CI tests over many conditioning sets	tests multiply when the graph is dense; early test errors propagate to later edges and directions
GES	compares scores of many edge additions/deletions at each step	with many variables score computations grow; greedy results depend on sample and search path

With maximal degree k, PC needs $O(p^{k+2})$ tests: polynomial in p, exponential in k.

- **MMHC**: CI tests restrict candidate edges, then score search;
- **FLOP**: efficient search over variable orders and parent sets.

5.1 MMHC

CI tests $\longrightarrow$ candidate skeleton $\longrightarrow$ score search $\longrightarrow$ DAG candidate

Stage 1: Max-Min Parents and Children finds candidate neighbours by CI tests. **Stage 2**: hill climbing on BIC within those edges. Edges dropped in stage 1 cannot return.

6 Comparison

	PC	GES	MMHC
approach	constraint (CI) based	score based	both, in sequence
search	removes edges when independent	changes the CPDAG to increase the score	restricts candidates by tests, then score search over DAGs
typical output	CPDAG	CPDAG	DAG candidate
key settings	independence test, significance level	score, complexity penalty	tests, significance level, score
sensitive to	test errors	score and search path	edges lost in stage 1

Summary

- Under the four assumptions, data identify the Markov equivalence class.
- PC: CI tests, v-structures from separating sets, Meek rules; correct with an oracle.
- BIC is decomposable and score equivalent; GES is consistent.
- MMHC combines both; stage-1 errors are irreversible.

7 Selected exercises

Motivation. BIC on one edge, soundness of the Meek rules, and the test count of PC.

Exercise 10.1 (BIC on one edge)

Two centered variables are observed n times, with sample variances s_X^2, s_Y^2 (divisor n) and sample correlation r, $|r| < 1$.
(a) Using the local scores of the BIC theorem, show that $S_{\mathrm{BIC}}(X \to Y) = S_{\mathrm{BIC}}(Y \to X)$ and that the edge beats the empty graph iff $-n \log(1 - r^2) > \log n$.
(b) For i.i.d. bivariate Gaussian data with correlation ρ, show that the probability that BIC chooses the correct adjacency tends to 1, both for $\rho = 0$ and for $\rho \neq 0$.

Solution [Supplement].

(a)

Step 1. The residual variance of Y on X is $s_Y^2 - s_{XY}^2/s_X^2 = s_Y^2(1 - r^2)$, so

$$S_{\mathrm{BIC}}(X \to Y) = -\frac{n}{2}\{\log s_X^2 + \log s_Y^2 + \log(1 - r^2)\} - \frac{3}{2}\log n + 2c,$$

symmetric in X, Y.

Step 2. Empty graph: $-\frac{n}{2}(\log s_X^2 + \log s_Y^2) - \log n + 2c$. Difference $-\frac{n}{2}\log(1 - r^2) - \frac{1}{2}\log n > 0 \iff -n \log(1 - r^2) > \log n$.

(b)

Step 3. ($\rho = 0$.) $\sqrt{n}\, r \xrightarrow{d} N(0, 1)$ (Ch. 8, $q = 0$), so $nr^2 = O_p(1)$, $r^2 \xrightarrow{p} 0$. For $0 \le u \le \frac{1}{2}$, $-\log(1 - u) = \sum_{k \ge 1} u^k/k \le u/(1 - u) \le 2u$, hence

$$P\{-n \log(1 - r^2) > \log n\} \le P\{2nr^2 > \log n\} + P\{r^2 > \tfrac{1}{2}\} \to 0.$$

Step 4. ($\rho \neq 0$.) $-\log(1 - r^2) \xrightarrow{p} -\log(1 - \rho^2) > 0$ and $n/\log n \to \infty$, so $-n \log(1 - r^2)/\log n \xrightarrow{p} \infty$: the edge is chosen with probability $\to 1$. ■

Exercise 10.2 (Soundness of the Meek rules)

Let $\mathcal{C}$ be a Markov equivalence class and P a partially directed graph with the common skeleton such that every directed edge of P has that orientation in every DAG of $\mathcal{C}$, and every v-structure of $\mathcal{C}$ appears in P with both edges directed. Show that if the premise of R1, R2 or R3 holds in P, then every DAG of $\mathcal{C}$ contains the edge that the rule orients.

Solution [Supplement]. Fix $G' \in \mathcal{C}$; each conclusion pair is adjacent, so we exclude the wrong orientation.

Step 1. (R1.) $G' \ni X \to Z$. If $Y \to Z$, then $X \to Z \leftarrow Y$ is a v-structure of $\mathcal{C}$, so $Y \to Z$ would be directed in P; but $Z - Y$ is undirected. So $Z \to Y$.

Step 2. (R2.) $Y \to X$ would give the cycle $X \to Z \to Y \to X$. So $X \to Y$.

Step 3. (R3.) Suppose $Y \to X$. Then $X \to Z_i$ gives the cycle $X \to Z_i \to Y \to X$, so $Z_i \to X$ $(i = 1, 2)$, and $Z_1 \to X \leftarrow Z_2$ is a v-structure that would be directed in P; but $X - Z_1$ is undirected. So $X \to Y$.

After PC's v-structure step the assumptions on P hold and each orientation preserves them, so all Meek orientations are correct. ■

Exercise 10.3 (How many tests PC needs)

Assume the conditions of the PC correctness theorem and let every node of the true DAG have at most k neighbours, $p \geq 3$. Show that the skeleton phase runs only the levels $\ell = 0, \ldots, k$ and performs at most $(k+1)p^{k+2}$ CI tests.

Solution [Supplement].

Step 1. A non-adjacent pair is separated by $\mathrm{pa}(X)$ and removed by level $|\mathrm{pa}(X)| \leq k$; true edges stay. So after level k the graph is the true skeleton.

Step 2. Then no node has more than k neighbours, so levels $\ell \geq k+1$ are never run.

Step 3. At level ℓ each ordered adjacent pair uses at most $\binom{|\mathrm{adj}(X_a)|-1}{\ell} \leq \binom{p-2}{\ell} \leq (p-2)^\ell$ sets, over at most $p(p-1)$ pairs:

$$p(p-1)\sum_{\ell=0}^{k}(p-2)^\ell \leq p^2(k+1)(p-2)^k \leq (k+1)p^{k+2}.$$

■